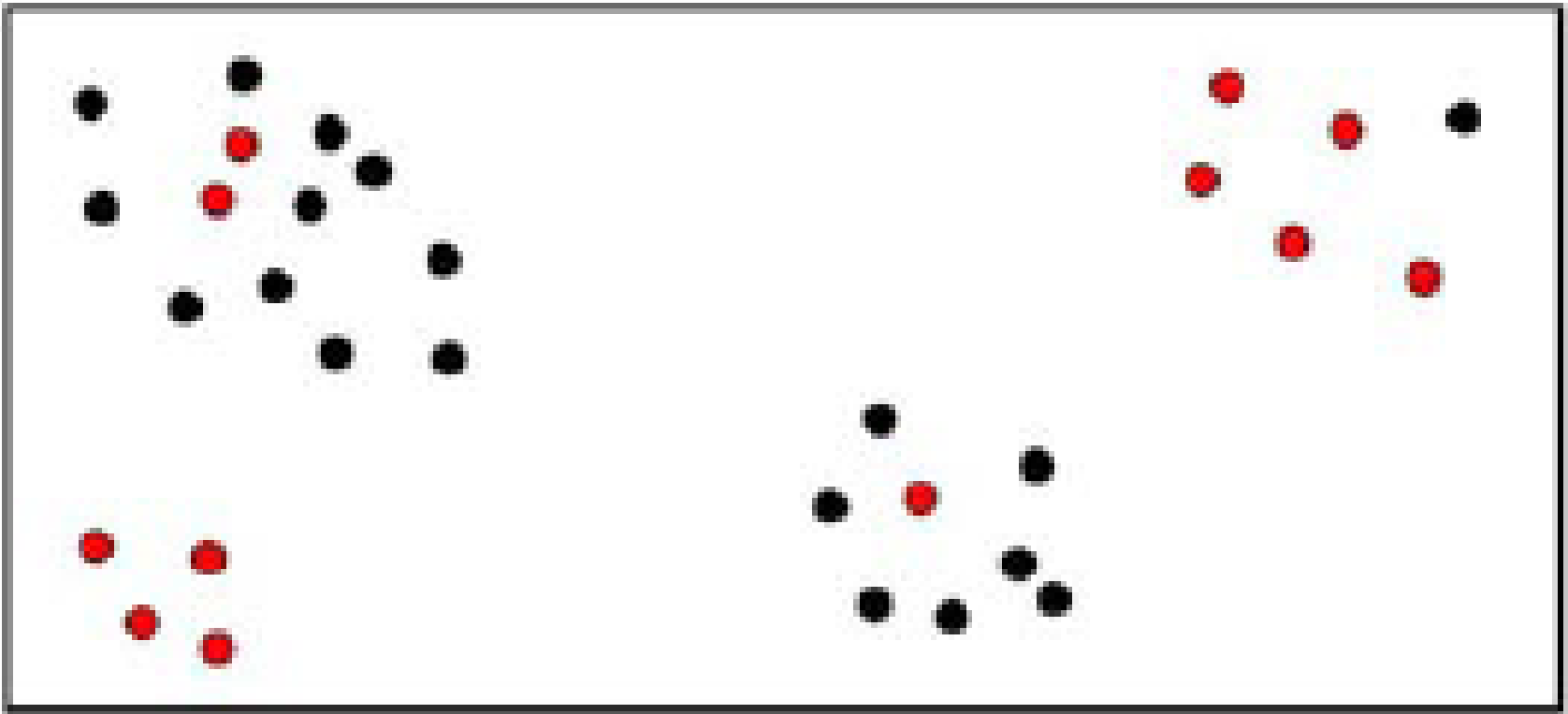
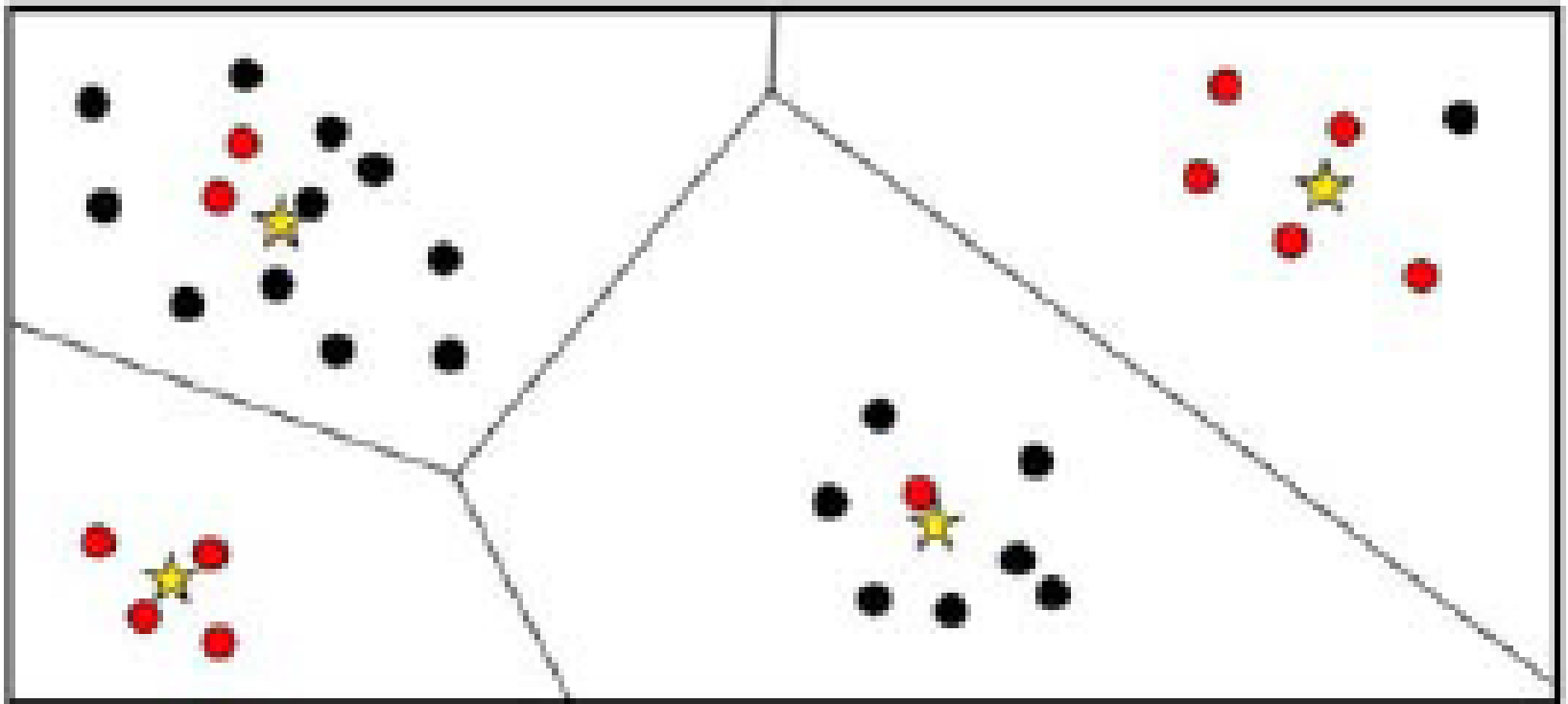


Makine Öğrenmesi - 5 (Machine Learning) Bölüm 18

Öbekleme (clustering)



Öbekleme (clustering)



Öbekleme (clustering)

- Bir öbek içindeki mesafeler az olmalı
- Öbekler arası mesafeler fazla olmalı
- Yine bir uzaklık ölçütü kullanmalıyız.

K-ortalama (K-means) öbekleme

- Tries to minimize

The diagram illustrates the K-means objective function, which aims to minimize the sum of squared distances between data points and their assigned cluster means. The equation is
$$\sum_{j=1}^k \sum_{i \in S_j} \|x^i - \mu_j\|^2$$
. Annotations include: a box labeled '# of clusters' pointing to the outer summation index k ; a box labeled 'elements of cluster j' pointing to the inner summation index $i \in S_j$; a box labeled 'mean of elts in cluster j' pointing to the mean μ_j ; and a box labeled 'squared dist from point to mean' pointing to the squared norm $\|x^i - \mu_j\|^2$.

- Only gets, greedily, to a local optimum

K-means algorithm

Choose k

Randomly choose k points C_j to be cluster centers

Loop

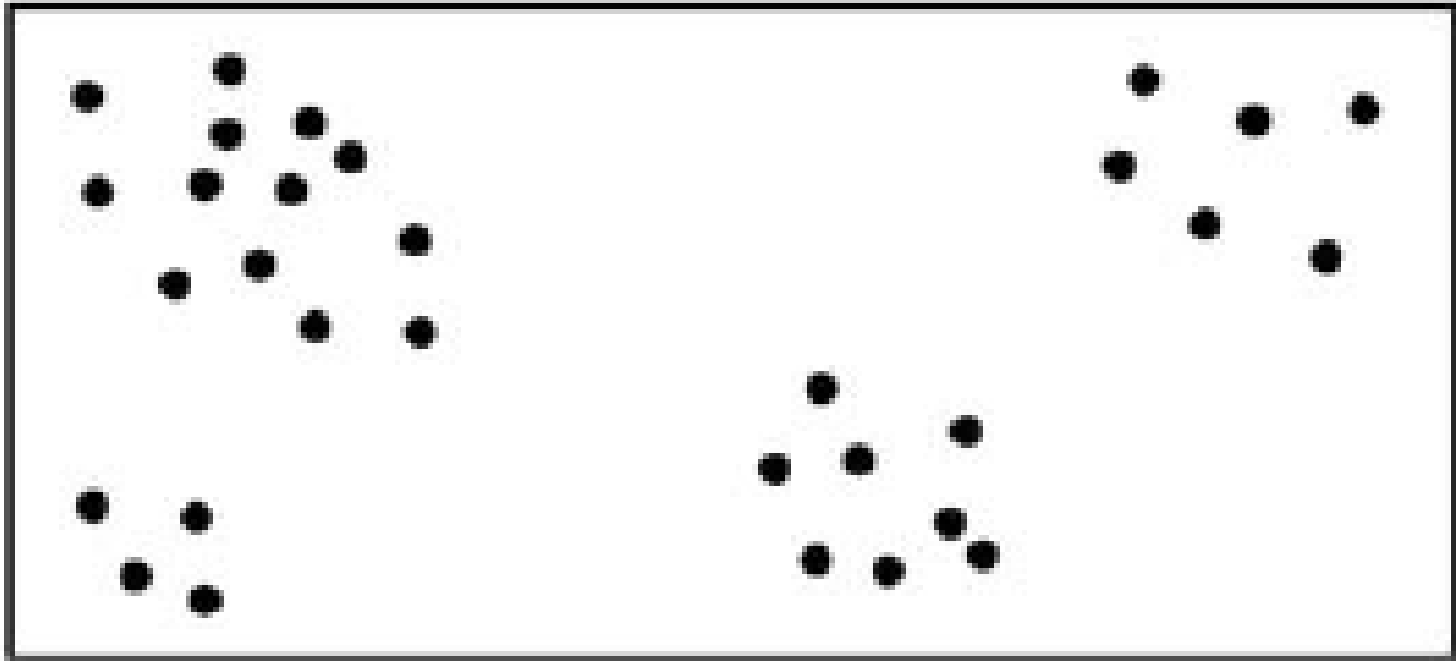
 Partition the data into k classes S_j according to which of the C_j they're closest to

 For each S_j , compute the mean of its elements and let that be the new cluster center

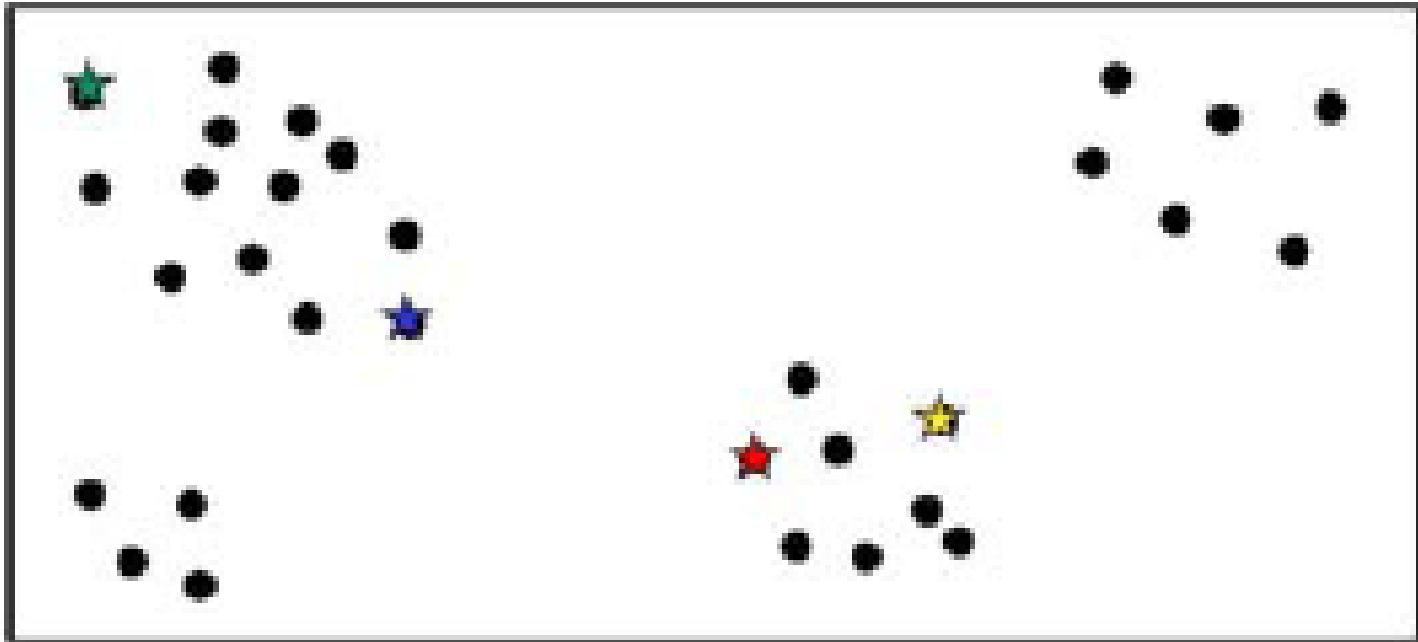
Stop when centers quit moving

- Guaranteed to terminate
- If a cluster becomes empty, re-initialize the center

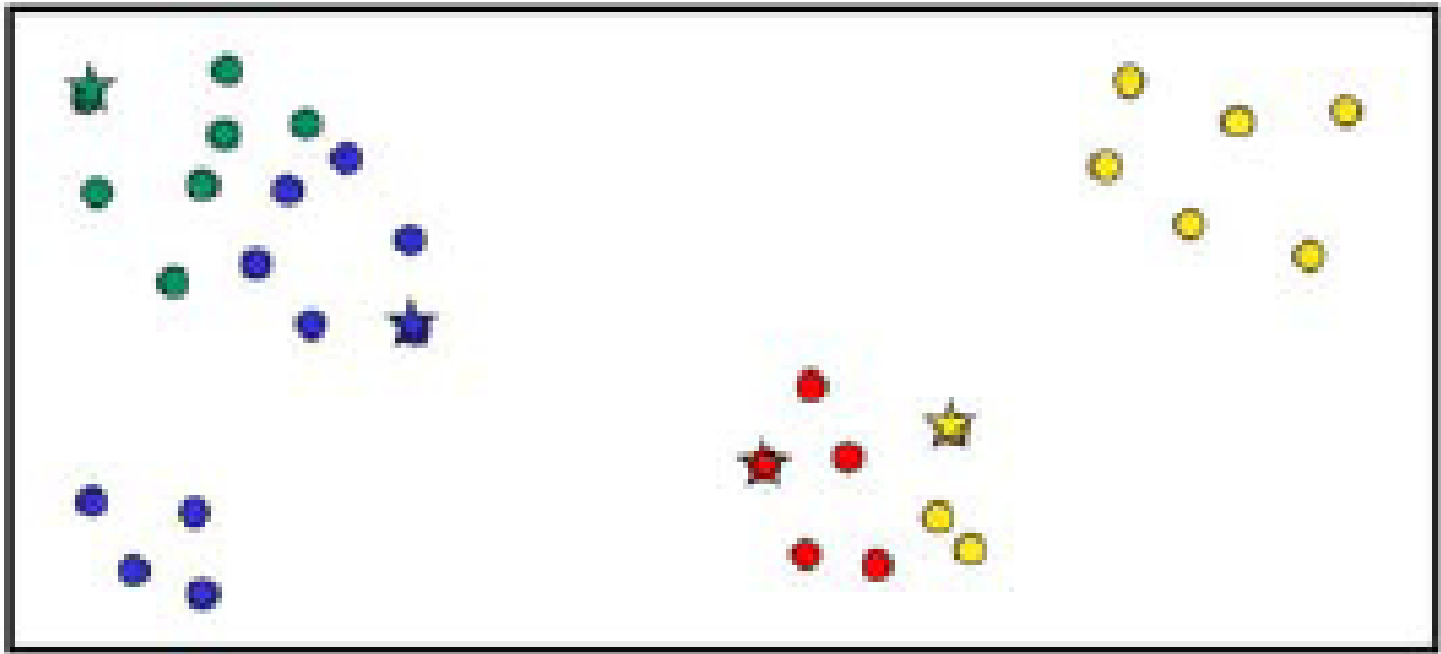
K-means örneği



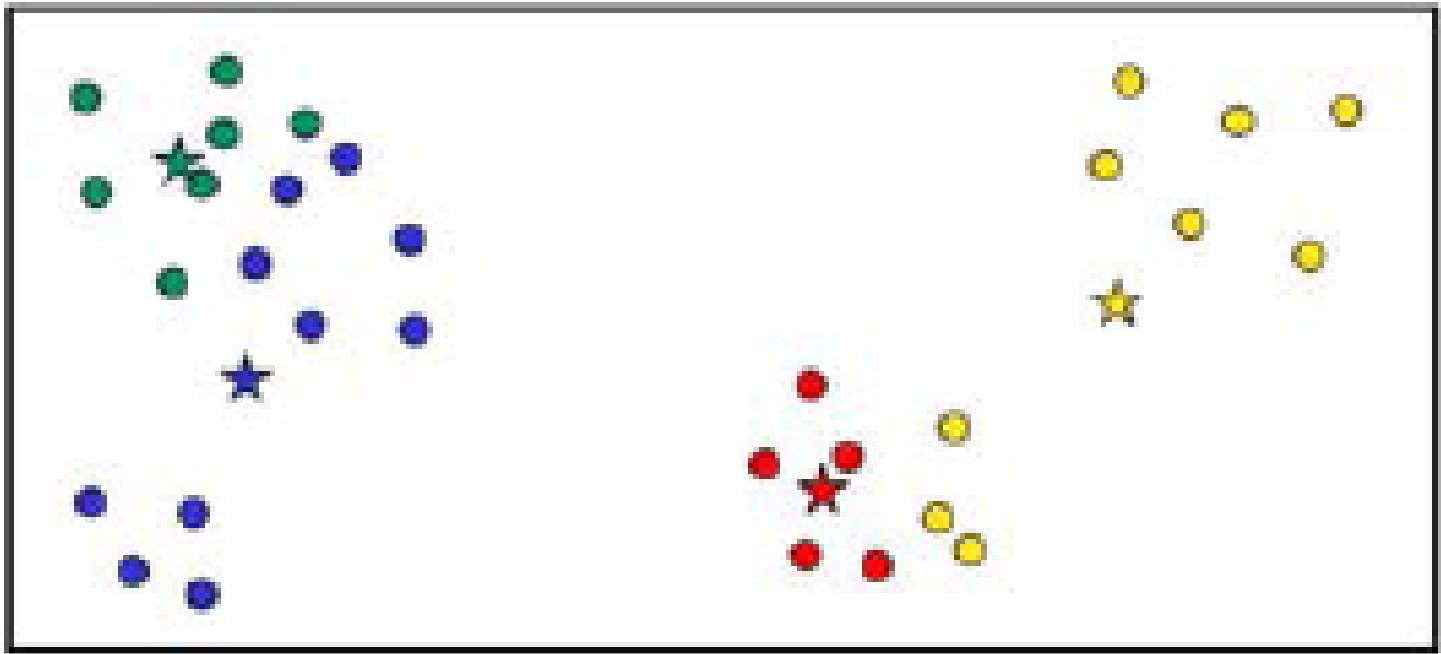
K-means örneği



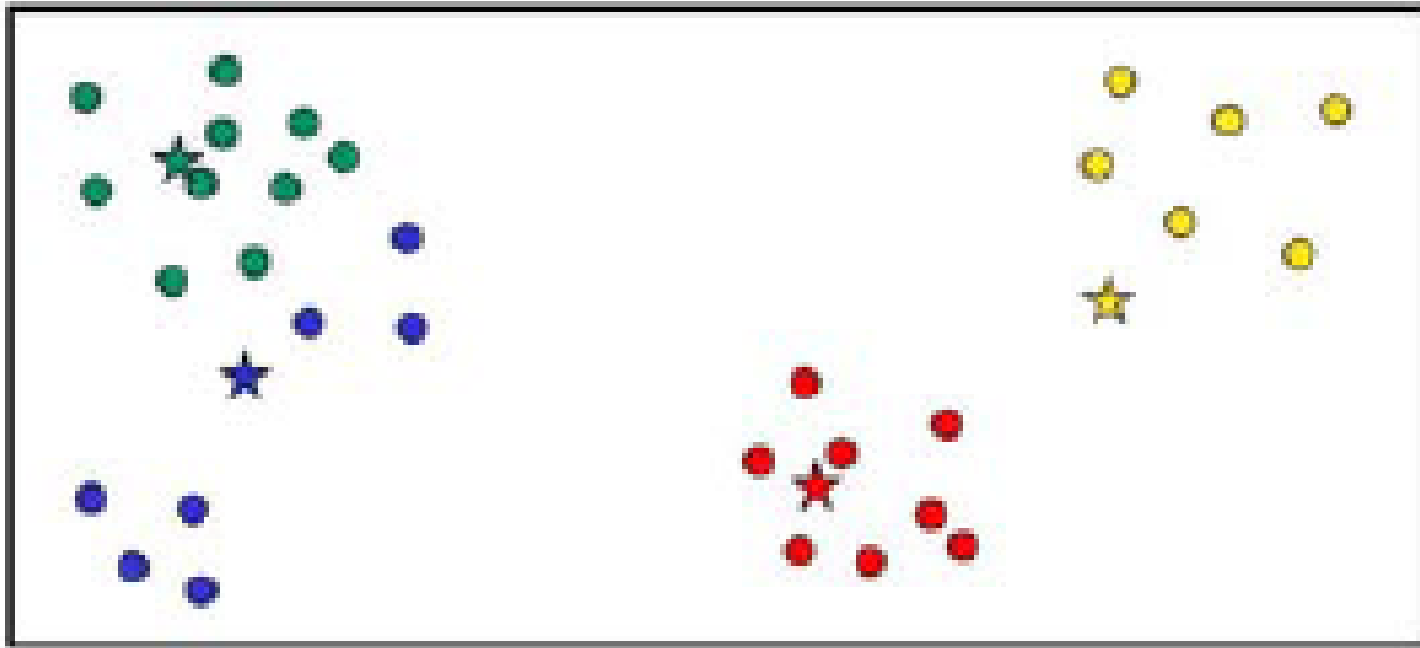
K-means örneği



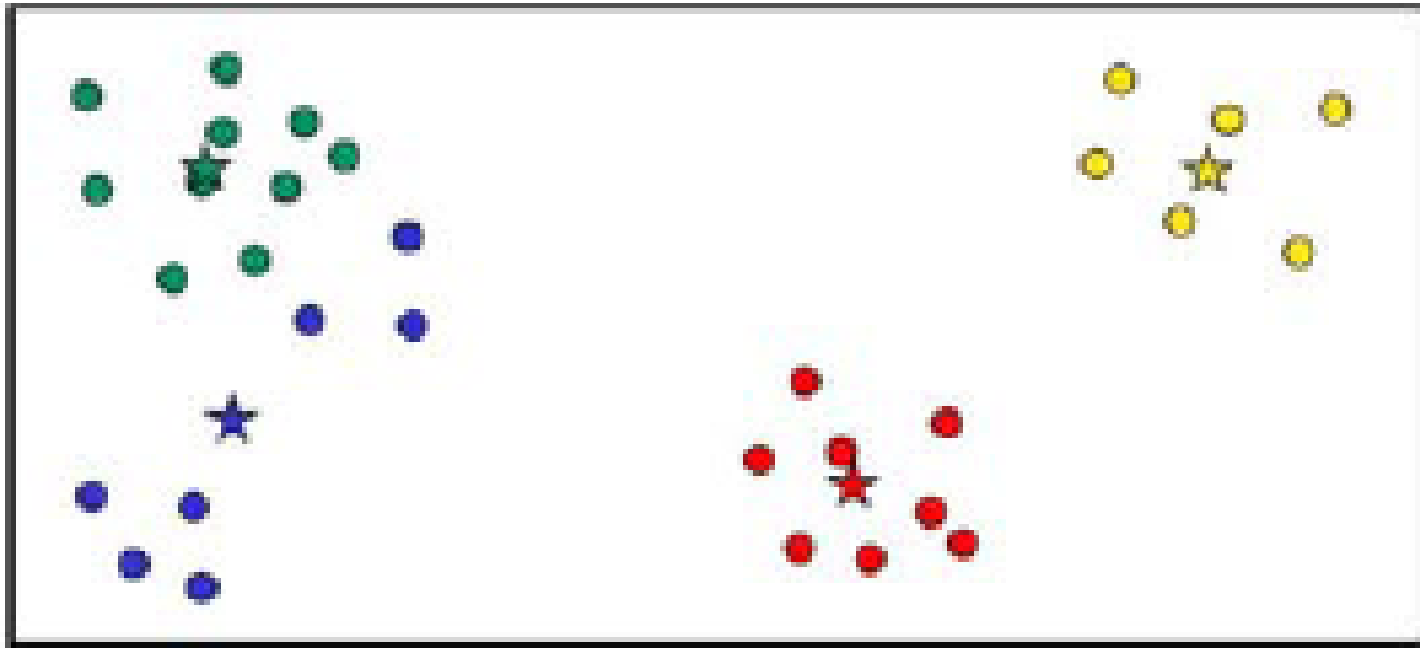
K-means örneği



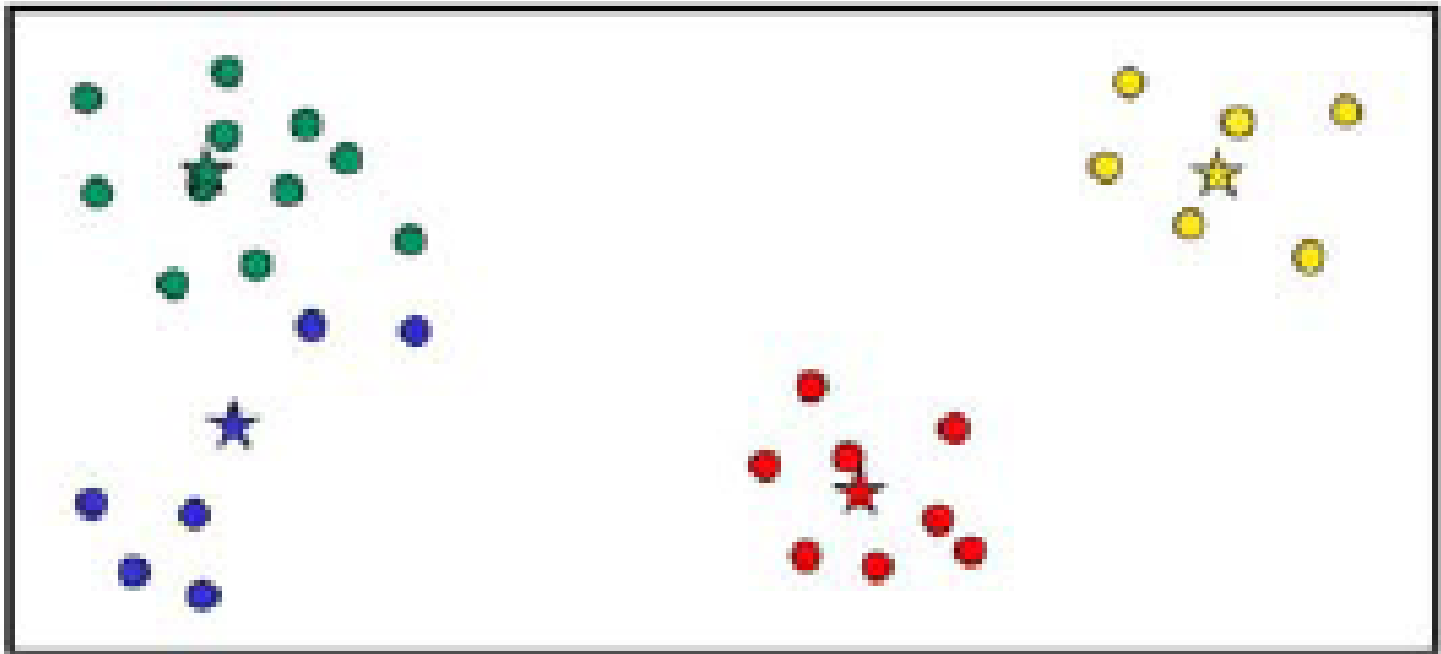
K-means örneği



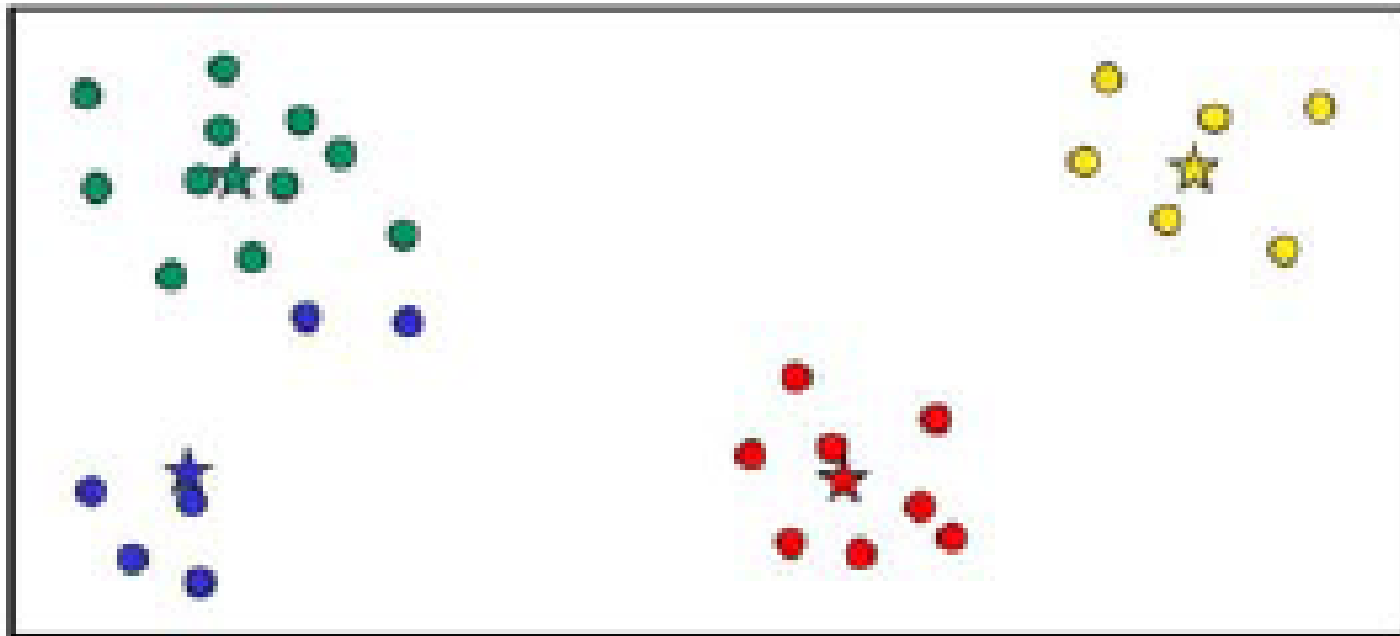
K-means örneği



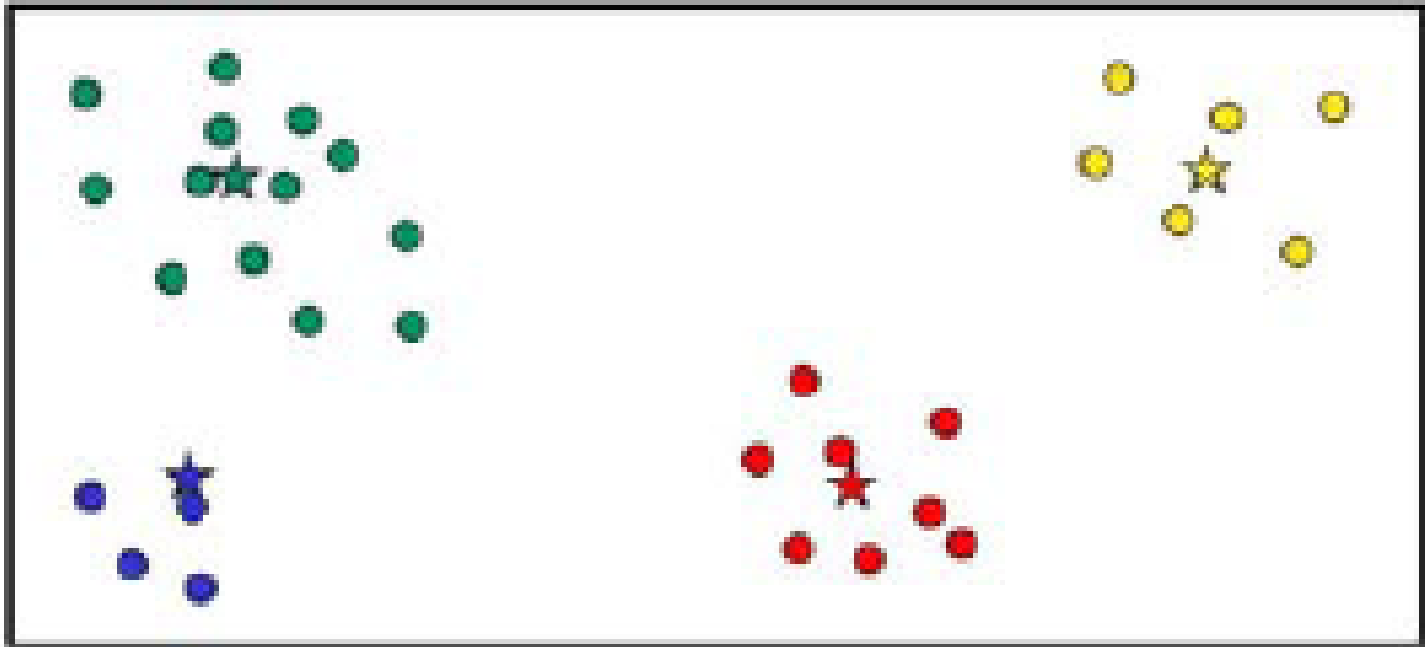
K-means örneği



K-means örneği



K-means örneği



K-means örneği

